# SIMATS ENGINEERING
## ASSESSMENT TOOL 1

**COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining**

---

**COURSE OUTCOME COVERED**

**CO5:** Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

---

**QUESTIONS: 05** **Total Marks:50**

Annexure A

Constraint-Based Problem Solving

| ***Criteria*** | Problem Understanding ***(2.5)*** | Constraint Analysis ***(2.5)*** | Solution Development ***(2.5)*** | Application of Concepts ***(1.5)*** | Justification ***(1)*** | ***Total(10)*** |
|---|---|---|---|---|---|---|
| *Weightage* | *25%* | 25% | 25% | 15% | 10% | *100%* |
| *Q1* | | | | | | |
| *Q2* | | | | | | |
| *Q3* | | | | | | |
| *Q4* | | | | | | |
| *Q5* | | | | | | |

---

***Code of Conduct:***

*I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

**RUBRICS FOR CALCULATION:**

## Constraint-Based Problem Solving Assessment Rubric

| Criteria | Weightage | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| Problem Understanding | 25% | Clearly defines the problem and identifies all relevant constraints accurately | Identifies most constraints with minor gaps | Partial understanding; misses key constraints | Fails to identify constraints correctly |
| Constraint Analysis | 25% | Analyzes constraints effectively and prioritizes them logically | Provides reasonable analysis with some prioritization | Limited analysis; weak prioritization | No meaningful analysis or prioritization |
| Solution Development | 25% | Develops optimal and feasible solutions satisfying all constraints | Develops workable solutions with minor limitations | Solutions partially satisfy constraints | Solutions do not meet constraints |
| Application of Concepts | 15% | Effectively applies appropriate concepts, methods, or tools | Applies concepts with minor errors | Limited application; requires guidance | Unable to apply relevant concepts |
| Justification | 10% | Provides strong justification for decisions with logical reasoning | Provides justification with some clarity | Weak or unclear justification | No justification provided |

**Annexure A**

## Q1. Dataset: Hospital Patient Records

| PID | Symptoms Observed |
|---|---|
| P1 | Fever, Cough |
| P2 | Fever, Headache |
| P3 | Cough, Headache |
| P4 | Fever, Cough, Headache |
| P5 | Fever, Fatigue |

**Question:**
Apply association rule mining with minimum support = 50% and confidence = 70%, under the constraint that all generated rules must include **"Fever"**. Identify the valid association rules and justify why they satisfy the given constraints.

---

## Q2. Dataset: Online Learning Platform

| Student ID | Courses Enrolled |
|---|---|
| S1 | Python, SQL |
| S2 | Python, Data Science |
| S3 | SQL, Machine Learning |
| S4 | Python, SQL, Data Science |
| S5 | Python, Machine Learning |

**Question:**
Using the dataset, generate frequent itemsets with minimum support = 40%, applying the constraint that each itemset must contain **at least two courses**. Explain how the constraint helps in pruning candidate itemsets during mining.

---

## Q3. Dataset: Library Borrowing Records

| Transaction ID | Books Borrowed |
|---|---|

| | |
|---|---|
| T1 | Data Mining, Python Programming |
| T2 | Data Mining, Database Systems |
| T3 | Python Programming, Database Systems |
| T4 | Data Mining, Python Programming, Database Systems |
| T5 | Data Mining, Artificial Intelligence |

**Question:**
Apply the Apriori algorithm with the constraint that the **maximum itemset size is 3**. Identify the frequent itemsets and explain how the constraint reduces computational complexity during candidate generation.

---

## Q4. Dataset: University Course Hierarchy

| Student ID | Courses Taken |
|---|---|
| S1 | Engineering, Computer Science, Machine Learning |
| S2 | Engineering, Information Technology |
| S3 | Engineering, Computer Science |
| S4 | Engineering, Computer Science, Machine Learning |
| S5 | Engineering, Information Technology |

**Question:**
Apply multilevel association rule mining with the constraint that rules should be generated only at the **"Computer Science"** level. Explain how the course hierarchy affects the discovery of association rules.

---

## Q5. Dataset: Insurance Customer Records

| CID | Age Group | Income Level | Policy Purchased |
|---|---|---|---|
| C1 | Young | High | Health Insurance |
| C2 | Middle | Medium | Vehicle Insurance |
| C3 | Young | Low | Travel Insurance |
| C4 | Senior | High | Health Insurance |
| C5 | Young | Medium | Vehicle Insurance |

**Question:**
Apply multidimensional association rule mining with the constraint **Age Group = "Young"**. Identify meaningful patterns involving income level and policy type, and explain how these patterns can support targeted marketing strategies.

## ANSWER

Q1 Answer:
Minimum support = 50% means an itemset must occur in at least 3 of 5 transactions. Fever occurs in P1, P2, P4, P5 = 80%; Cough occurs with Fever in P1, P4 = 40%; Headache occurs with Fever in P2, P4 = 40%; Fatigue occurs with Fever in P5 = 20%.
Therefore, no 2-itemset containing Fever reaches 50% support. Hence, no association rule containing Fever satisfies both minimum support 50% and confidence 70%. The constraint eliminates all candidate rules.

## ANSWER

Q2 Answer:
Minimum support = 40% means an itemset must occur in at least 2 of 5 transactions. With the constraint that every itemset has at least two courses, the frequent 2-itemsets are:
{Python, SQL}: 2/5 = 40%
{Python, Data Science}: 2/5 = 40%
{Python, Machine Learning}: 1/5 = 20% (not frequent)
{SQL, Machine Learning}: 1/5 = 20% (not frequent)
{SQL, Data Science}: 1/5 = 20% (not frequent)
{Data Science, Machine Learning}: 0%.
No 3-itemset reaches 40%. Thus the frequent constrained itemsets are {Python, SQL} and {Python, Data Science}. The size constraint prunes all single-course itemsets and reduces candidate generation.

## ANSWER

Q3 Answer:
Using Apriori, minimum support is not explicitly stated, so the frequent itemsets are identified using the given transactions and the usual count-based candidates. With the maximum itemset size constrained to 3, candidate generation stops after 3-itemsets.
1-itemsets: Data Mining (4), Python Programming (2), Database Systems (3), Artificial Intelligence (1).
Frequent 2-itemsets at support count >= 2:
{Data Mining, Python Programming} (2)
{Data Mining, Database Systems} (2)
{Python Programming, Database Systems} (2)
Frequent 3-itemset:
{Data Mining, Python Programming, Database Systems} (1), so it is not frequent at count >=2.
The maximum-size constraint prevents generation of itemsets larger than 3, reducing the number of candidates, support-counting operations, memory use, and runtime.

## ANSWER

Q4 Answer:
At the Computer Science level, consider only rules involving Computer Science and its co-occurring course at the specified level. Computer Science occurs in S1, S3 and S4 = 3/5 = 60%.
Machine Learning occurs with Computer Science in S1 and S4 = 2/5 = 40%.
Therefore the meaningful association is:
Computer Science -> Machine Learning, with support = 2/5 = 40% and confidence = 2/3 = 66.7%.
The hierarchy allows mining at a selected abstraction level instead of mixing Engineering, Computer Science and Machine Learning indiscriminately. Restricting the mining to Computer Science reduces irrelevant candidate rules and produces more focused knowledge.

## ANSWER

Q5 Answer:
Apply the constraint Age Group = Young. The relevant records are C1, C3 and C5:
C1: High income -> Health Insurance
C3: Low income -> Travel Insurance
C5: Medium income -> Vehicle Insurance
Among Young customers, High, Low and Medium income each occur once (33.3%), and each policy type also occurs once (33.3%). Thus there is no strong repeated income-policy association in this small dataset. However, the multidimensional pattern shows that policy preference varies across income levels: High-income young customers purchased Health Insurance, Low-income customers purchased Travel Insurance, and Medium-income customers purchased Vehicle Insurance. This can support targeted marketing by tailoring offers to income segments, while recognizing that the sample is too small to establish a statistically strong rule.